Simulation and Analysis of a Multi-type Galton–Watson Process

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1 Introduction

The Galton–Watson (GW) process is a classical stochastic model used to describe the evolution of a population across discrete generations. Starting from an initial population, each individual independently generates a random number of offspring according to a fixed probability distribution. The model is widely used in population dynamics, branching processes, and probabilistic modeling.

In this laboratory work, we study a *multi-type Galton–Watson process*, where individuals are divided into different classes. A simulation-based approach is used to empirically estimate the asymptotic extinction probability, and the numerical results are compared with theoretical predictions. Confidence intervals are also computed to quantify the variability of the empirical estimates.

2 Theoretical Background

2.1 Classical Galton–Watson Process

Let X_n denote the number of individuals in generation n. The evolution of the population is described by

$$X_n = \sum_{j=1}^{X_{n-1}} Y_j, \tag{1}$$

where the random variables Y_j represent the number of offspring generated by individual j and are assumed to be independent and identically distributed.

The process becomes extinct when $X_n = 0$ for some n, in which case no further generations can be produced.

In the Poisson case, where $Y_j \sim \text{Poisson}(m)$, the extinction probability q satisfies the fixed-point equation

$$q = e^{-m(1-q)}. \tag{2}$$

If $m \leq 1$, extinction occurs almost surely, whereas for $m > 1$ extinction occurs with probability $q < 1$.

2.2 Multi-type Galton–Watson Process

The Galton–Watson process can be extended to the *multi-type* case by allowing individuals to belong to different classes. Let C denote the number of types and define

$$Z_n = (Z_{n,1}, \ldots, Z_{n,C}), \tag{3}$$

where $Z_{n,c}$ is the number of individuals of type c in generation n.

Each individual of type c produces a random vector of offspring

$$Y_c^{(v)} = (Y_{c,1}^{(v)}, \ldots, Y_{c,C}^{(v)}), \tag{4}$$

where $Y_{c,c'}^{(v)}$ denotes the number of offspring of type c' produced by individual v of type c.

The population evolves according to

$$Z_{n+1} = \sum_{c=1}^{C} \sum_{v=1}^{Z_{n,c}} Y_c^{(v)}. \tag{5}$$

2.3 Poisson Multi-type Model

In this work, we consider the Poisson multi-type case, where

$$Y_{c,c'}^{(v)} \sim \text{Poisson}(m_{c,c'}). \tag{6}$$

The process is fully characterized by the mean offspring matrix

$$M = (m_{c,c'}). \tag{7}$$

The extinction probability is now a vector

$$q = (q_1, \ldots, q_C), \tag{8}$$

where q_c is the probability of extinction starting from a single individual of type c.

For Poisson offspring distributions, the extinction probabilities satisfy the system

$$q_c = \exp\left(\sum_{c'=1}^{C} m_{c,c'}(q_{c'} - 1)\right), \quad c = 1, \ldots, C. \tag{9}$$

2.4 Supercriticality Condition

In the classical GW process, the process is supercritical if $m > 1$. In the multi-type case, this condition generalizes to

$$\rho(M) > 1, \tag{10}$$

where $\rho(M)$ denotes the spectral radius (largest eigenvalue) of the matrix M.

2.5 Confidence Intervals

The extinction probability is estimated empirically via Monte Carlo simulations. Given B independent estimates $\hat{q}_1, \dots, \hat{q}_B$, the mean extinction probability is

$$\bar{q} = \frac{1}{B} \sum_{i=1}^{B} \hat{q}_i. \tag{11}$$

By the Central Limit Theorem, a 95% confidence interval is approximated by

$$\bar{q} \pm 1.96 \frac{s}{\sqrt{B}}, \tag{12}$$

where s is the sample standard deviation of the estimates.

3 Model Specification

We focus on the case $C = 2$, with mean offspring matrix

$$M(\alpha) = \begin{pmatrix} \frac{6}{7}\alpha & \frac{2}{7}\alpha \\ \frac{2}{7}\alpha & \frac{3}{7}\alpha \end{pmatrix}, \tag{13}$$

where $\alpha \in \{0.9, 0.95, 1.0, 1.05, 1.1\}$.

For this matrix, the spectral radius satisfies

$$\rho(M(\alpha)) = \alpha. \tag{14}$$

Therefore, the process is subcritical for $\alpha < 1$, critical for $\alpha = 1$, and supercritical for $\alpha > 1$.

4 Implementation and Numerical Results

4.1 Implementation Details

The multi-type Galton–Watson process described in the previous sections was implemented in Python using a simulation-based approach. At each generation, the population is represented by a vector

$$Z_n = (Z_{n,1}, Z_{n,2}),$$

where $Z_{n,c}$ denotes the number of individuals of type c at generation n.

For a fixed value of the parameter α, the mean offspring matrix is defined as

$$M(\alpha) = \begin{pmatrix} \frac{6}{7}\alpha & \frac{2}{7}\alpha \\ \frac{2}{7}\alpha & \frac{3}{7}\alpha \end{pmatrix}.$$

For each individual of type c, an offspring vector is generated by sampling independent Poisson random variables with means given by the c-th row of $M(\alpha)$.

The simulation starts from a single individual, either of type 1 or type 2. At each generation, offspring vectors are generated for all individuals and summed to obtain the population of the next generation. The simulation stops when one of the following conditions is met:

- extinction occurs, i.e. $Z_n = (0, 0)$,
- the maximum number of generations is reached,
- the total population size exceeds a predefined threshold, in which case the process is considered to have survived.

To obtain reliable empirical estimates, each branching process simulation is run for at most 200 generations and extinction is declared when the population reaches zero. A truncation threshold of 500 individuals is introduced: whenever the total population exceeds this value, the process is classified as surviving and the simulation is stopped early. This criterion improves numerical stability without biasing extinction estimates.

For each value of the parameter α, extinction probabilities and survival curves are computed from 5,000 independent realizations. To quantify statistical uncertainty, confidence intervals are constructed using a batch approach: $B = 20$ independent repetitions are performed, each consisting of $n = 200$ simulations. The resulting empirical extinction probabilities are averaged across batches and compared with the corresponding theoretical predictions.

4.2 Survival Curves

For each value of $\alpha \in \{0.9, 0.95, 1.0, 1.05, 1.1\}$, survival curves were computed as

$$P(\text{survival up to generation } t) = 1 - P(\text{extinction by generation } t).$$

Figure 1 shows the survival curves when the process starts from a single individual of type 1, while Figure 2 shows the corresponding curves when starting from a single individual of type 2.

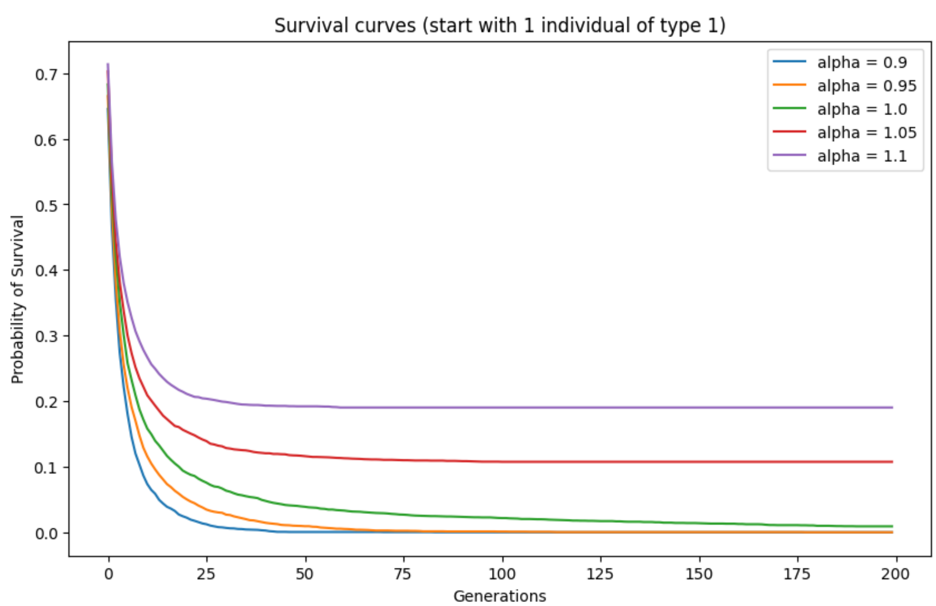

Figure 1: Survival curves for the multi-type Poisson Galton–Watson process starting from one individual of type 1.

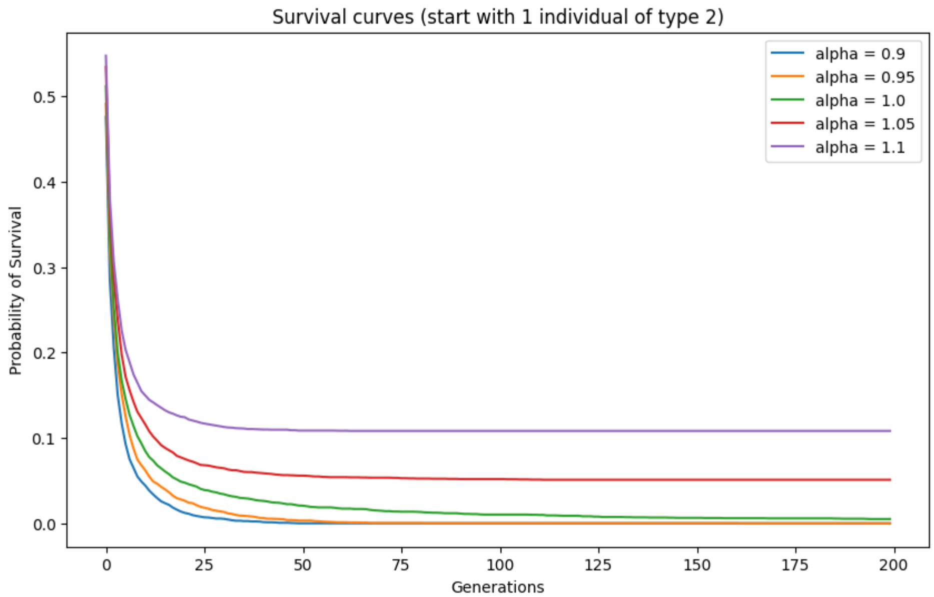

Figure 2: Survival curves for the multi-type Poisson Galton–Watson process starting from one individual of type 2.

The numerical results clearly reflect the theoretical classification of the process:

- **Subcritical regime ($\alpha = 0.9, 0.95$).** The survival probability rapidly decreases to zero within a small number of generations, indicating that extinction occurs almost surely.
- **Critical regime ($\alpha = 1.0$).** The survival probability decreases more slowly and remains slightly above zero even after many generations. This behavior is consistent with the theoretical result that extinction is certain, but convergence is slow and affected by the finite simulation horizon.
- **Supercritical regime ($\alpha = 1.05, 1.1$).** The survival curves stabilize at positive values, indicating a strictly positive probability of survival. Moreover, larger values of α correspond to higher long-term survival probabilities.

A comparison between Figures 1 and 2 shows that starting from a type 1 individual generally leads to slightly higher survival probabilities, due to the larger expected number of offspring produced by type 1 individuals.

4.3 Extinction Probabilities and Confidence Intervals

The asymptotic extinction probability was estimated empirically as the proportion of simulations that eventually became extinct. To assess the statistical variability of the estimates, the computation was repeated multiple times and 95% confidence intervals were constructed using a normal approximation.

Table 1 reports, for each value of α, the empirical extinction probability (mean), the corresponding confidence interval, and the theoretical extinction probability obtained by solving the fixed-point equations of the multi-type Galton–Watson process.

Table 1: Empirical and theoretical extinction probabilities for the multi-type Poisson Galton–Watson process.

α	q_1^{emp}	95% CI	q_1^{theo}	q_2^{emp}	q_2^{theo}
0.9	1.000	(1.000, 1.000)	1.000	1.000	1.000
0.95	1.000	(1.000, 1.000)	1.000	1.000	1.000
1.0	0.988	(0.983, 0.992)	0.999	0.996	0.999
1.05	0.898	(0.889, 0.906)	0.896	0.949	0.946
1.1	0.799	(0.787, 0.811)	0.806	0.893	0.896

The empirical extinction probabilities are in very good agreement with the theoretical values. In the subcritical regime, extinction is observed with probability one, as predicted by theory. In the critical case, the extinction probability is slightly below one due to the finite number of generations considered in the simulations. In the supercritical regime, extinction probabilities are strictly less than one and decrease as α increases.

4.4 Supercriticality Analysis

The supercriticality condition was investigated by computing the spectral radius of the mean offspring matrix $M(\alpha)$. The numerical results show that

$$\rho(M(\alpha)) = \alpha.$$

Therefore, the process is supercritical if and only if $\alpha > 1$. This theoretical condition perfectly matches the empirical findings, since positive survival probabilities are observed only for $\alpha > 1$.
The theoretical extinction probabilities are computed by solving the fixed-point equation

$$q = \exp(M(q - \mathbf{1}))$$

which is implemented via monotone iteration in the function. Empirical extinction probabilities are estimated through Monte Carlo simulations.

In the general multi-type setting, the process is supercritical if and only if the spectral radius $\rho(M)$ is larger than one. This condition is implemented in the code by computing the eigenvalues of M using the function `spectral_radius`. The numerical results show that $\rho(M(\alpha)) = \alpha$, so that the process is subcritical for $\alpha < 1$, critical for $\alpha = 1$, and supercritical for $\alpha > 1$. This theoretical classification is fully consistent with the empirical survival curves and extinction probabilities observed in the simulations.

4.5 Discussion

The numerical experiments confirm the theoretical properties of multi-type Galton–Watson processes. In particular, they highlight the central role of the spectral radius of the mean offspring matrix in determining the long-term behavior of the process. The close agreement between empirical estimates and theoretical predictions validates the simulation approach and demonstrates the effectiveness of the implemented model.

Simulation of a Multi-Group Epidemic Model with Public Health Interventions and Vaccination

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Abstract

This report presents the formulation and simulation of a Multi-Group SEIQRS-V epidemiological model designed to capture population inhomogeneity, spatial heterogeneity, and the effects of public health policies. We partition the population into three distinct groups (Youth, Adults, Elderly) with varying risks of exposure and fatality. The model incorporates mechanisms for the isolation of asymptomatic individuals, limited natural immunity, and dynamic vaccination strategies. Through computational simulation using the Euler method, we analyze the interplay between vaccination priorities (Targeted vs. Mass) and Non-Pharmaceutical Interventions (NPIs) such as lockdowns. Results demonstrate that while vaccination is the ultimate solution, NPIs are critical in delaying the epidemic peak to allow vaccination campaigns to reach vulnerable groups, particularly under high-contagion scenarios.

1 Introduction

The objective of this Lab is to simulate a realistic epidemic scenario where the population is not a homogeneous block. Real-world epidemics are driven by complex interactions between different demographic groups who share different contact patterns and biological vulnerabilities. Furthermore, the management of an epidemic relies on a combination of pharmaceutical interventions (vaccines) and non-pharmaceutical interventions (isolation, lockdowns). This report derives the Mean-Field Equations for a stratified SEIQRS-V model and analyzes the outcomes of eight distinct scenarios ranging from uncontrolled spread to coordinated containment strategies.

2 Theoretical Framework

2.1 Model Definition

We define a deterministic compartmental model where the population is divided into $M = 3$ groups: **Youth** (G), **Adults** (L) and **Elderly** (A). For each group k, the total population N_k is constant (ignoring natural births/deaths unrelated to the epidemic). The compartments are defined as follows:

- S_k: Susceptible individuals.
- E_k: Exposed individuals (infected but not yet infectious; latent period).

- $I_{a,k}$: Asymptomatic Infectious individuals (undetected).
- $I_{s,k}$: Symptomatic Infectious individuals (undetected).
- Q_k: Quarantined/Isolated individuals (detected and removed from the transmission pool).
- R_k: Recovered individuals (temporarily immune).
- V_k: Vaccinated individuals.
- D_k: Deceased due to the disease.

To visualize the transitions between these states, we refer to the standard compartmental flow.

2.2 Heterogeneity and Force of Infection

The core of the multi-group model is the **Contact Matrix** C_{kj}, which represents the average number of contacts an individual in group k has with individuals in group j. This captures spatial and social heterogeneity.

The Force of Infection acting on group k, denoted as $\lambda_k(t)$, depends on the prevalence of infectious individuals in all groups j, weighted by the contact matrix and the transmission probability β:

$$\lambda_k(t) = \beta \sum_{j=1}^{M} C_{kj} \cdot \frac{I_{a,j} + I_{s,j}}{N_j} \tag{1}$$

Note that individuals in compartment Q are assumed to be effectively isolated and do not contribute to λ.

2.3 Mean-Field Differential Equations

The dynamics of the epidemic for each group k are governed by the following system of non-linear differential equations:

$$\frac{dS_k}{dt} = -\lambda_k S_k + \omega R_k - \nu_k(t) \tag{2}$$
$$\frac{dE_k}{dt} = \lambda_k S_k - \sigma E_k \tag{3}$$
$$\frac{dI_{a,k}}{dt} = (1 - p_k)\sigma E_k - \gamma I_{a,k} - \eta_a I_{a,k} \tag{4}$$
$$\frac{dI_{s,k}}{dt} = p_k \sigma E_k - \gamma I_{s,k} - \mu_k I_{s,k} - \eta_s I_{s,k} \tag{5}$$
$$\frac{dQ_k}{dt} = \eta_a I_{a,k} + \eta_s I_{s,k} - \gamma Q_k - \mu_k Q_k \tag{6}$$
$$\frac{dR_k}{dt} = \gamma(I_{a,k} + I_{s,k} + Q_k) - \omega R_k \tag{7}$$
$$\frac{dV_k}{dt} = \nu_k(t) \tag{8}$$
$$\frac{dD_k}{dt} = \mu_k(I_{s,k} + Q_k) \tag{9}$$

2.4 Parameter Definitions

- β: Transmission coefficient (modified by NPIs).
- σ: Rate of progression from Exposed to Infectious (1/latency).
- γ: Recovery rate (1/infectious period).
- ω: Rate of waning immunity (flow from $R \to S$).
- p_k: Probability of developing symptoms for group k.
- μ_k: Daily mortality rate for symptomatic individuals in group k.
- η_a, η_s: Detection and isolation rates for asymptomatic and symptomatic cases, respectively.
- $\nu_k(t)$: Dynamic vaccination rate for group k, subject to supply constraints.

3 Computational Implementation

3.1 Model Architecture

To address the complexity of a heterogeneous epidemic, we formulated a Multi-Group SEIQRS-V model (Susceptible - Exposed - Infected Asymptomatic - Infected Symptomatic - Quarantined - Recovered - Vaccinated). The population of $N = 100,000$ individuals is stratified into three distinct demographic groups to capture varying social behaviors and biological risks:

- **Group 0 (Youth):** Characterized by high contact rates but a low probability of developing severe symptoms.
- **Group 1 (Adults):** Represents the core workforce with high mobility and medium clinical risk.
- **Group 2 (Elderly):** Characterized by lower social mobility but an extremely high risk of mortality upon infection.

Initial conditions and model parameters. At the initial time, the population is almost entirely susceptible. To initiate the epidemic dynamics, a small deterministic infection seed is introduced by moving 50 individuals from the susceptible compartment to the asymptomatic infectious compartment within the adult group. This corresponds to approximately 0.05% of the total population. All remaining compartments are initialized to zero.

The epidemic dynamics are governed by a set of biological and epidemiological parameters. The transition from the exposed to the infectious state is regulated by the parameter $\sigma = 1/3$, corresponding to an average latent period of approximately 3 days. Recovery from infection occurs at rate $\gamma = 1/10$, yielding an average infectious duration of about 10 days. Temporary post-infection immunity is assumed, with immunity waning on a time scale of approximately 180 days, modeled through the parameter $\omega = 1/180$.

The probability of developing symptomatic infection is group-dependent and reflects increasing clinical severity with age. Specifically, it is set to $p_0 = 0.4$ for the youth group, $p_1 = 0.7$ for adults, and $p_2 = 0.95$ for the elderly. Detection and isolation of infected individuals occur at different rates depending on clinical presentation, with asymptomatic individuals detected at rate $\eta_a = 0.1$, and symptomatic individuals detected more rapidly at rate $\eta_s = 0.4$.

Mortality is assumed to affect only symptomatic and quarantined individuals and is strongly age-dependent. The mortality parameters are group-dependent and fixed throughout the simulations. They are set to $\mu_0 = 0.001$ for youth, $\mu_1 = 0.002$ for adults, and $\mu_2 = 0.05$ for the elderly, reflecting the strongly increasing fatality risk with age.

Finally, the overall transmission intensity is controlled by the parameter β, whose value varies across scenarios, and by a structured contact matrix that is modulated by non-pharmaceutical intervention factors to represent mobility restrictions and social distancing measures.

Unlike simple theoretical models, this algorithm incorporates realistic logistical constraints and policy interventions.

Vaccination Mechanism Vaccination is modeled not merely as a continuous rate, but as a resource-constrained process. Each day, a limited number of individuals (controlled by the parameter *vax_rate*) are moved from the susceptible compartment to the vaccinated compartment, following a predefined priority order (*vax_strategy*). Vaccination therefore reduces infections indirectly by decreasing the size of the susceptible population, while it does not directly affect either the mortality rate or the duration of the disease.

To evaluate different public health approaches, we implemented two distinct allocation strategies:

- **Targeted Strategy (Mortality Control):** Vaccination doses are allocated according to a fixed priority order (Elderly $\rightarrow$ Adults $\rightarrow$ Youth). This strategy aims to minimize fatalities by acting on groups with the highest mortality rates μ_k, effectively decoupling infection prevalence from death counts.
- **Mass Strategy (Transmission Control):** Vaccination doses are allocated according to the priority order (Adults $\rightarrow$ Youth $\rightarrow$ Elderly). The objective is to reduce the force of infection λ_k by immunizing individuals with the highest contact rates, thereby suppressing viral circulation.

Both strategies employ a **soft-switch mechanism**: as the susceptible population of a priority group approaches zero, the vaccination flow is automatically redirected to the next group. This ensures that no vaccine doses are wasted and that state variables remain non-negative at all times.

Non-Pharmaceutical Interventions (NPIs) Lockdowns are implemented as a direct modulation of the **contact matrix** C. Specifically, NPIs act through a scalar parameter $\alpha \in (0, 1]$, referred to as the *NPI factor*, which rescales the baseline contact matrix as

$$C = \alpha\, C_{\text{base}}.$$

While the biological parameters of the virus (such as the transmission rate β) remain unchanged, the NPI factor reduces the frequency of effective contacts between demographic groups, thereby lowering the force of infection.

- **No Lockdown:** $\alpha = 1.0$, corresponding to normal social activity and 100% of baseline contacts.
- **Soft Lockdown:** $\alpha = 0.7$, simulating moderate social distancing measures, with a 30% reduction in effective contacts.

- **Hard Lockdown:** $\alpha = 0.4$, simulating strict stay-at-home policies, with a 60% reduction in effective contacts.

This mechanism acts as a "brake" on the Force of Infection (λ), delaying the epidemic peak to buy time for the vaccination campaign to take effect.

Isolation and Quarantine Distinct from general lockdowns, the model explicitly includes a **Quarantine compartment** (Q_k), which represents the capacity of the system to detect and isolate infectious individuals through testing, contact tracing, or hospitalization.

Isolation is governed by two detection parameters:

- η_a: the detection and isolation rate of **asymptomatic infectious** individuals ($I_{a,k}$),
- η_s: the detection and isolation rate of **symptomatic infectious** individuals ($I_{s,k}$).

In the simulations, these rates are set to $\eta_a = 0.1$ and $\eta_s = 0.4$, reflecting a higher probability of detecting symptomatic cases. Detected individuals are transferred from $I_{a,k}$ or $I_{s,k}$ to the quarantine compartment Q_k.

Individuals in Q_k are completely removed from the transmission process: they do not contribute to the force of infection λ_k, unlike undiagnosed infectious individuals. However, quarantined individuals may still recover at rate γ or die at a group-dependent mortality rate μ_k.

This mechanism allows the model to distinguish between population-wide mobility restrictions (NPIs acting on the contact matrix C) and targeted public-health interventions that directly remove infectious individuals from circulation.

3.2 Implementation

The system of differential equations was solved using the **LSODA algorithm** (via Python's `scipy.integrate.odeint`), which provides superior numerical stability and adaptive time-stepping compared to simple fixed-step methods.

To incorporate discrete vaccination constraints into this continuous solver, we implemented a Robust Priority Logic:

1. **Continuous Vaccination Rate:** The daily vaccine supply (V_{max}) is distributed dynamically among groups based on a specified strategy (e.g., *Targeted* vs. *Mass*).
2. **Smooth Saturation:** A soft switching function was applied to the vaccination rate $\nu(t)$. As the susceptible population of a priority group approaches zero, the vaccination rate smoothly decays to prevent numerical artifacts (such as negative populations) and automatically redirects resources to the next priority group.
3. **Non-Negativity Enforcement:** State variables are continuously monitored to ensure biological feasibility throughout the integration interval.

Calibration Parameters The following parameters are modified across the scenarios to simulate different epidemic contexts and government responses:

- **Transmission Rate (β):** Represents the aggressiveness of the viral strain.

- **High Contagion ($\beta = 0.6$):** Simulates a highly infectious variant (e.g., Omicron-like transmissibility).
- **Low Contagion ($\beta = 0.3$):** Simulates a moderate strain or seasonal influenza.

- **Vaccination Campaign (ν):** Defined by both speed and priority.
  - **Rate:** We test a **Weak** rollout (200 doses/day, simulating logistical bottlenecks) versus a **Strong** rollout (800 doses/day, simulating an efficient mass campaign).
  - **Strategy:** *Targeted* prioritizes the Elderly (high mortality risk), while *Mass* prioritizes the Workforce/Youth (high transmission risk).

- **Non-Pharmaceutical Interventions (NPI Factor):** A scalar multiplier applied to the Contact Matrix C to simulate social restrictions.
  - **No Lockdown (1.0):** Normal life, 100% of baseline contacts.
  - **Soft Lockdown (0.7):** Moderate restrictions (e.g., masks, remote working), reducing contacts to 70%.
  - **Hard Lockdown (0.4):** Strict stay-at-home orders, reducing contacts to 40%.

Output Metrics (Table Columns) The performance of each scenario is summarized using four key cumulative indicators at $t = 250$ days:

- **Final S (%):** The percentage of the population that remains Susceptible at the end of the simulation. A high value indicates successful containment (the virus did not reach everyone).
- **Peak Infected (%):** The maximum percentage of the population simultaneously infected ($I_{active} + Q$) at the height of the wave. This is the critical metric for assessing hospital saturation ("flattening the curve").
- **Final V (%):** The total percentage of the population successfully vaccinated by the end of the period.
- **Final Deaths (%):** The cumulative mortality rate. In a high-mortality context ($\mu_{elderly} = 5\%$), this is the primary metric for defining the success or failure of a strategy.

4 Visual Overview of Dynamics

The simulation results were categorized into four macro-scenarios to isolate the effects of intervention intensity and strategic prioritization.

4.1 Full Compartment Dynamics and Visualization Choices

In principle, the correct visualization of this multi-group epidemic model should include the full set of compartments describing the population state, namely: Susceptible (S_k), Exposed (E_k), Asymptomatic Infectious ($I_{a,k}$), Symptomatic Infectious ($I_{s,k}$), Quarantined (Q_k), Recovered (R_k), Vaccinated (V_k) and Deceased (D_k). Figure 1 shows the complete time evolution of all these compartments for the baseline scenario, corresponding to high contagion, no vaccination, and no non-pharmaceutical interventions.

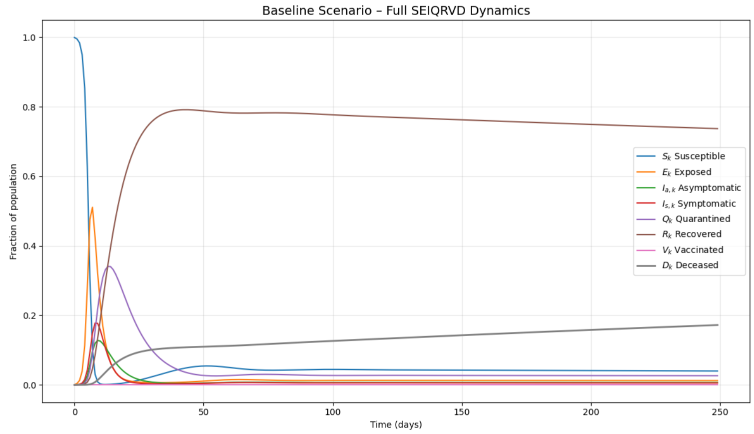

Figure 1: Baseline scenario: full SEIQRVD dynamics aggregated over all population groups. The plot includes all epidemiological compartments of the model.

While this representation is theoretically complete, it is not always the most effective for comparative analysis across multiple scenarios. Several compartments (such as Exposed, Asymptomatic, Symptomatic, and Quarantined) mainly serve as intermediate states and tend to overlap visually, making interpretation more difficult when many simulations are displayed simultaneously.

For this reason, in the remainder of the analysis we focus on a reduced set of key compartments that best summarize the macroscopic epidemic dynamics: Susceptible (S), Active Infected (defined as the sum of infectious and quarantined individuals), Recovered (R), Vaccinated (V) and Deceased (D). All other compartments are still fully simulated within the model, but they are omitted from the plots for clarity and readability. This simplified visualization allows for a clearer comparison of epidemic severity, control effectiveness, and long-term outcomes across different policy scenarios.

4.2 Macro-Scenario 1: Baseline and Weak Interventions

This set analyzes the "worst-case" conditions where the virus ($\beta = 0.6$) spreads with minimal opposition.

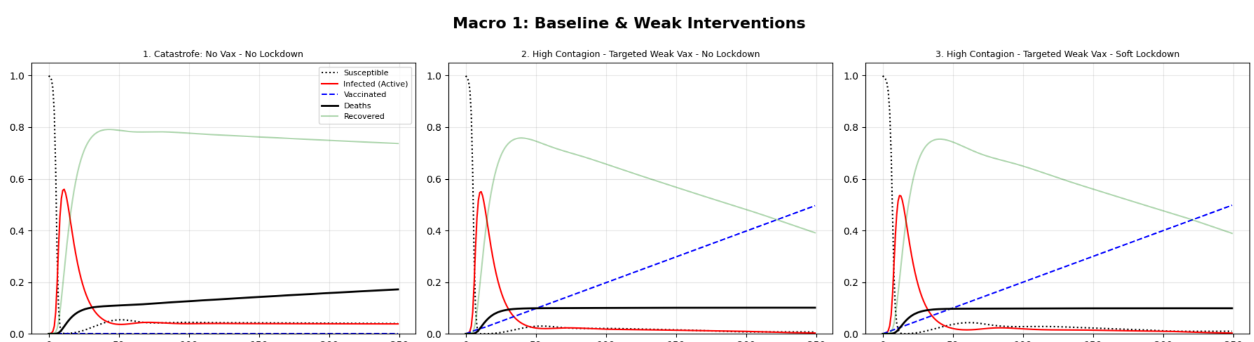

Figure 2: Group A: Dynamics of Unmitigated Spread (Scen 1) vs. Weak Vaccination (Scen 2, 3).

Scenario 1 (Catastrophe) serves as the baseline for total failure. In the complete absence of vaccines and with normal social interactions maintained ($NPI = 1.0$), the virus encounters no resistance. The resulting dynamic shows an immediate and violent infection peak that overwhelms the population before any natural immunity can develop, leading to the highest mortality rate among all simulations.

In **Scenario 2**, an attempt is made to mitigate the disaster by introducing a vaccination campaign targeted at the elderly, but with limited logistical capacity (*Weak rate*: 200 doses/day). However, by keeping social contacts at 100%, the speed of viral propagation drastically outpaces the speed of immunization. The result highlights that a slow vaccine rollout, without the support of restrictions, arrives "too late" to save the majority of the vulnerable population.

Finally, **Scenario 3** demonstrates the concept of synergy. While maintaining the slow vaccination rate, a "Soft Lockdown" is applied, reducing contacts by 30% ($NPI = 0.7$). Graphically, we observe that this restriction does not eliminate the virus but significantly slows its race ("flattening the curve"). This buys valuable time for the weak vaccination campaign to cover a larger portion of the at-risk population before they become infected.

4.3 Macro-Scenario 2: Impact of Strong Interventions

This group contrasts the effectiveness of pharmaceutical interventions (Vaccines) against non-pharmaceutical ones (Lockdowns).

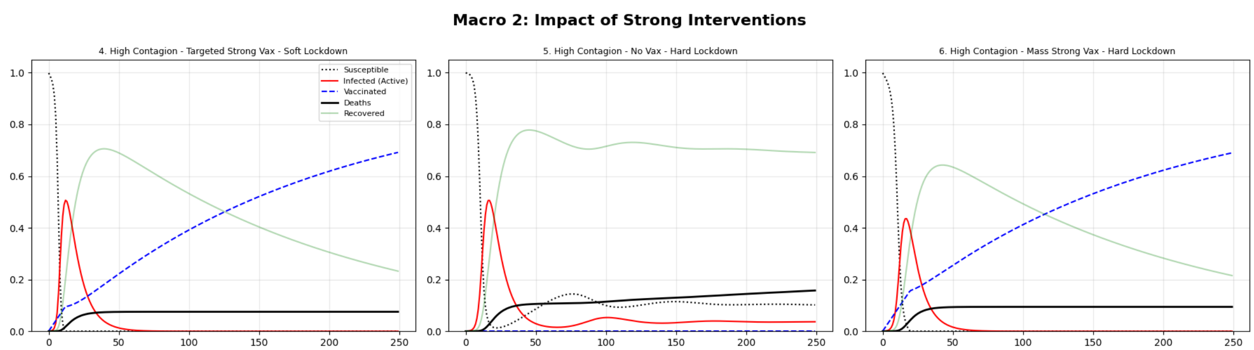

Figure 3: Group B: Strong Vaccination + Soft NPI (Scen 4) vs. Hard Lockdowns (Scen 5, 6).

Scenario 4 represents the ideal realistic management strategy. By combining a Targeted Vaccination with moderate social restrictions ("Soft Lockdown", $NPI = 0.7$), we achieve a drastic reduction in mortality. The partial reduction in contacts slows the viral spread just enough to allow the strong vaccination drive to immunize the elderly before the infection wave reaches them, effectively decoupling the infection rate from the death rate.

In contrast, **Scenario 5** tests the limits of physical isolation alone. Implementing a "Hard Lockdown" ($NPI = 0.4$) without any vaccination reveals a critical systemic weakness. While the infection curve is heavily suppressed, the population remains biologically susceptible. This demonstrates that strict NPIs act merely as a "pause button"; without the exit strategy provided by vaccines, the virus continues to circulate (albeit slowly) among a vulnerable population, resulting in significant mortality despite the economic cost of the lockdown.

Finally, **Scenario 6** applies maximum force: a Hard Lockdown combined with Strong Vaccination. However, the vaccination strategy here is "Mass" (prioritizing the workforce/youth). While viral transmission is almost completely suffocated, the mortality outcome is not optimal compared to targeted approaches. This highlights a paradox: suppressing the spread via lockdown and mass vaccination is less efficient at saving lives than simply prioritizing the elderly, as the vulnerable group remains exposed while the low-risk population is immunized.

4.4 Macro-Scenario 3: Strategy Comparison (Mass vs. Targeted)

Here we compare the ethical and practical trade-off: should we stop the spreaders (Youth/Adults) or shield the vulnerable (Elderly)?

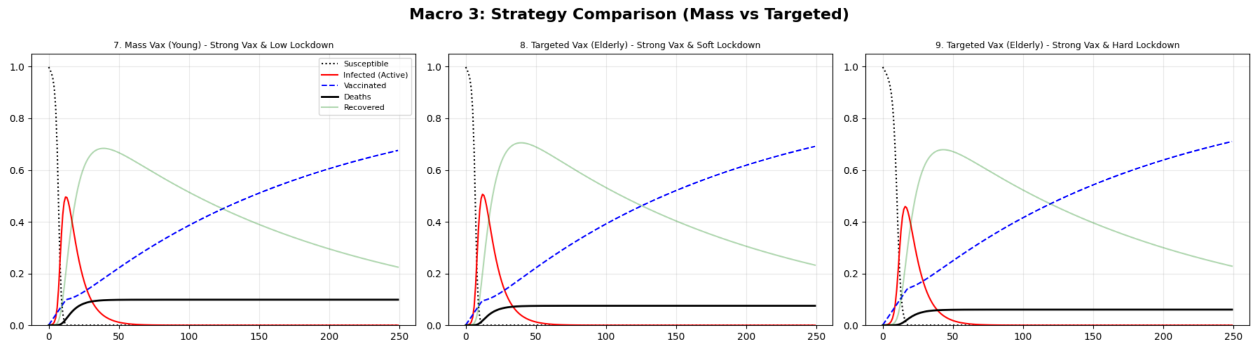

Figure 4: Group C: Mass Vaccination (Scen 7) vs. Targeted Vaccination (Scen 8, 9).

Scenario 7 tests the "Mass Vaccination" approach combined with a Soft Lockdown. By prioritizing the Youth and Adults—who possess the highest contact rates—this strategy effectively aims to break the chain of transmission. While this results in a notable suppression of the infection peak, it fails to minimize mortality efficiently. The virus, though slowed, still reaches the unvaccinated Elderly population, leading to a higher death toll compared to targeted approaches.

Scenario 8 utilizes the exact same resources (Strong Rate, Soft Lockdown) but shifts the priority to "Targeted Vaccination" (Elderly First). The difference in outcome is substantial. Although the virus circulates more freely among the low-risk youth, the high-risk demographic is immunized early in the curve. This strategy mathematically demonstrates that in a high-mortality context, shielding the vulnerable yields a far better survival rate than attempting to suppress general transmission.

Finally, **Scenario 9** represents the absolute optimal outcome of the entire simulation. By combining the most efficient priority queue (Targeted) with the strictest containment measure (Hard Lockdown, $NPI = 0.4$), the system achieves the "Maximum." The Hard Lockdown suppresses the force of infection (λ) to such an extent that the vaccination campaign has ample time to cover not just the elderly, but also significant portions of the adult population before they are exposed, resulting in the lowest cumulative mortality of all tested scenarios.

4.5 Macro-Scenario 4: Low Contagion Dynamics

This final group validates the model sensitivity under less aggressive viral conditions ($\beta = 0.3$), simulating a scenario akin to a seasonal influenza or a less transmissible variant, to determine if strict measures are still necessary.

Scenario 10 represents a highly controlled environment: low viral transmissibility combined with a Targeted Vaccination campaign and a Soft Lockdown. The results indicate that when the force of infection is naturally lower, even a "Weak" vaccination rate is sufficient to suppress the epidemic almost entirely. The curve is flattened to such an extent that the healthcare system faces minimal pressure, and mortality is kept at near-nominal levels.

Scenario 11 serves as a control test: Low Contagion with Soft Lockdown, but *no vaccines*. Despite the lower β, the absence of immunity means the virus eventually finds its way through the population. While

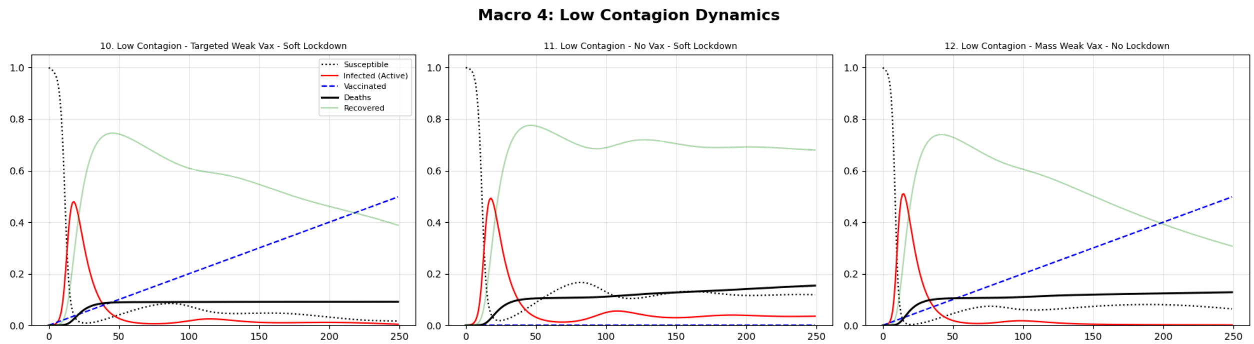

Figure 5: Group D: Dynamics under Low Contagion ($\beta = 0.3$).

the peak is nowhere near the catastrophic levels of Group A, the cumulative mortality remains significant over time. This reinforces the finding that NPIs alone—even against a milder virus—are not a permanent solution without acquired immunity.

Finally, **Scenario 12** explores the possibility of returning to "normal life" ($NPI = 1.0$). Here, we use a Weak Vaccination rate with a "Mass" strategy (no prioritization) against the low-contagion virus. Unlike in high-contagion scenarios where this approach failed, here it proves moderately effective. Because the virus spreads slowly, the "Weak" vaccine rollout is able to keep pace with the infection rate. This leads to the conclusion that the strictness of required interventions (Lockdowns and Prioritization) is directly dependent on the pathogen's intrinsic transmissibility (β); lower-risk viruses allow for more flexible management strategies.

5 Results and Analysis

This section analyzes the quantitative outcomes of the twelve simulated scenarios (Table 1). The simulations were conducted under a "High Mortality" calibration to stress-test the effectiveness of various intervention strategies. The primary metrics for evaluation are the Final Cumulative Deaths (as a percentage of the total population), the Peak Infection Rate, and the Final Vaccination coverage.

Table 1: Simulation Results: SIRVD Metrics for 12 Scenarios (Updated Data)

Scenario	**Final S (%)**	**Peak Infected (%)**	**Final V (%)**	**Final Deaths (%)**
1. Catastrofe: No Vax - No Lockdown	4.0%	56.0%	0.0%	17.22%
2. High Contagion - Targeted Weak Vax - No Lockdown	0.7%	55.0%	49.6%	10.13%
3. High Contagion - Targeted Weak Vax - Soft Lockdown	1.0%	53.6%	49.8%	9.91%
4. High Contagion - Targeted Strong Vax - Soft Lockdown	0.0%	50.7%	69.2%	7.54%
5. High Contagion - No Vax - Hard Lockdown	10.2%	50.6%	0.0%	15.77%
6. High Contagion - Mass Strong Vax - Hard Lockdown	0.0%	43.6%	69.0%	9.44%
7. Mass Vax (Giovani) - Strong Mass Vax and Low Lockdown	0.0%	49.6%	67.6%	9.90%
8. Targeted Vax (Anziani) - Strong Targeted Vax and Soft Lockdown	0.0%	50.7%	69.2%	7.54%
9. Targeted Vax (Anziani) - Strong Targeted Vax and Hard Lockdown	0.0%	45.9%	71.1%	6.07%
10. Low Contagion, Targeted Weak Vax, Low Lockdown	1.7%	48.0%	49.8%	9.16%
11. Low Contagion, No Vax, Low Lockdown	11.9%	49.3%	0.0%	15.43%
12. Low Contagion, Mass Weak Vax, No Lockdown	6.4%	51.0%	49.8%	12.84%

A critical evaluation of the simulation data reveals three fundamental insights regarding epidemic management. First, **Non-Pharmaceutical Interventions** (NPIs) are insufficient when used in isolation. Comparing the baseline **Scenario 1** (17.22% deaths) with **Scenario 5** (Hard Lockdown, 15.77% deaths) indicates that strict mobility restrictions alone reduced final mortality by only $\sim$1.5%. This confirms that NPIs function primarily as a delay mechanism; without a pharmaceutical exit strategy, they fail to prevent mass casualties in a high-mortality context. In fact, even a "Weak" vaccination campaign without lockdowns (**Scenario 2**, 10.13% deaths) proved significantly more effective than a strict lockdown without vaccines.

Second, the data strongly favors a **Targeted Vaccination** strategy over Mass Vaccination. Although prioritizing the workforce in **Scenario 6** achieved a lower peak infection rate (43.6%) by suppressing spreaders, it resulted in over **50% higher mortality** (9.44%) compared to **Scenario 9** (Targeted Vaccination, 6.07% deaths). This mathematically demonstrates that prioritizing the demographic with the highest Case Fatality Rate (μ) is far more efficient for minimizing total deaths than attempting to suppress general transmission.

Ultimately, the most effective strategy was the synergistic combination found in **Scenario 9**, where Targeted Strong Vaccination coupled with a Hard Lockdown reduced baseline deaths by nearly 65%.

6 Conclusion

Based on the simulation results, we draw the following conclusions regarding epidemic management:

1. **Vaccination is the Primary Solution:** Immunization is the only effective method to drastically reduce mortality. Even a weak vaccination campaign significantly outperforms strict lockdowns in the long run.

2. **Lockdowns are Supportive, Not Curative:** Restrictions on mobility are necessary to flatten the curve and prevent healthcare collapse, but they cannot stop the epidemic indefinitely. They must be used as a bridge to vaccination.

3. **Targeted Strategies Save Lives:** Prioritizing the elderly yields significantly better mortality outcomes than mass vaccination strategies, particularly when the virus has a high fatality rate in specific age groups.

Random Graph Generation and Dynamic Processes

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Abstract

This report presents an experimental and theoretical study of random graph models and dynamical processes defined on them. Using the `NetworkX` library and efficient spatial indexing via `scipy.spatial.cKDTree`, we analyze large-scale random networks. We study probabilistic graphs $G(n, p)$, focusing on the emergence of a giant connected component and the connectivity threshold through Galton–Watson branching process arguments, and we investigate diameter scaling in the connected regime. An efficient linear-time algorithm for generating sparse random graphs is implemented and empirically validated. The analysis is extended to random geometric graphs $G(n, r)$, highlighting the impact of spatial constraints on connectivity and diameter. Finally, we simulate a Majority Model to show how network structure amplifies small initial biases, leading to rapid global consensus.

1 Random Graphs $G(n, p)$

We consider random graphs with n nodes, where each possible edge between two distinct nodes is independently added with probability p. Despite its simplicity, this model exhibits rich structural properties and sharp phase transitions as the parameter p varies.

All simulations were implemented using the `NetworkX` library, which provides efficient adjacency-list representations and optimized routines for graph generation and traversal.

Our analysis focuses on:

- the size of the largest and second-largest connected components;
- the conditions under which the graph becomes fully connected;
- the behavior of the graph diameter when connectivity is achieved.

1.1 Connected Components and Branching Process Theory

To understand the emergence of large connected components, we rely on the theory of Galton–Watson (GW) branching processes. Starting from a randomly chosen node, the exploration of its neighbors and their neighbors can be approximated by a branching process.

When $p = a/n$, the degree distribution converges to a Poisson distribution with mean a. The behavior of the graph is therefore governed by the offspring mean of the associated branching process:

- **Subcritical regime ($a < 1$):** The branching process dies out almost surely. As a consequence, all connected components remain small, typically of size $O(\log n)$.

- **Supercritical regime ($a > 1$):** The branching process survives with positive probability. In a finite graph, this corresponds to the emergence of a *giant connected component* whose size scales linearly with n.

The diameter of a graph G is defined as

$$\mathrm{diam}(G) = \max_{u,v \in V} d(u, v),$$

where $d(u, v)$ is the length of the shortest path between nodes u and v. This definition is meaningful only when the graph is connected. If the graph is disconnected, there exist pairs of nodes for which $d(u, v) = \infty$, and therefore the diameter is not defined.

1.2 Experiment 1: Linear Regime $p = a/n$

1.2.1 Implementation and Setup

We generated random graphs with

$$n \in \{10^4, 10^5, 10^6\},$$

and values of a close to the critical threshold. For each parameter combination, at least 10 independent graph instances were generated.

Connected components were identified using breadth-first search (BFS). The components were sorted by size to extract the largest and second-largest connected components.

Linear Case Results (p=a/n)

n	a	Largest (mean)	Second (mean)	Connected (%)	Diameter
10000	0.9	128.5	95.1	0.0%	N/A
10000	0.95	240.1	153.7	0.0%	N/A
10000	0.98	392.9	131.1	0.0%	N/A
10000	1.0	272.1	158.4	0.0%	N/A
10000	1.02	406.9	233.6	0.0%	N/A
10000	1.05	671.9	312.3	0.0%	N/A
10000	1.1	1877.6	153.2	0.0%	N/A
100000	0.9	371.2	223.9	0.0%	N/A
100000	0.95	544.9	422.8	0.0%	N/A
100000	0.98	829.3	644.2	0.0%	N/A
100000	1.0	2200.6	779.7	0.0%	N/A
100000	1.02	4765.6	850.1	0.0%	N/A
100000	1.05	8035.2	661.3	0.0%	N/A
100000	1.1	17874.5	322.9	0.0%	N/A
1000000	0.9	450.0	400.0	0.0%	N/A
1000000	0.95	1421.5	933.7	0.0%	N/A
1000000	0.98	2501.5	1962.1	0.0%	N/A
1000000	1.0	7717.1	4180.3	0.0%	N/A
1000000	1.02	34284.3	4867.5	0.0%	N/A
1000000	1.05	96198.5	1191.4	0.0%	N/A
1000000	1.1	177987.8	561.9	0.0%	N/A

Figure 1: Experimental results for the linear regime ($p = a/n$).

1.2.2 Results and Discussion

The experimental results clearly confirm the phase transition predicted by GW theory. For $a < 1$, all connected components remain small. As a approaches 1, the second-largest component increases in size, indicating strong competition between clusters.

Once $a > 1$, a giant connected component emerges abruptly. For example, when $n = 10^6$ and $a = 1.1$, the largest component contains approximately 18% of the nodes, while the second-largest component collapses to a negligible size.

Despite the emergence of a giant component, the graph remains disconnected with high probability in the linear regime. A linear number of isolated nodes persists for all values of a, implying that the probability of full connectivity tends to zero as $n \to \infty$. This explains why the percentage of connected instances observed in the experiments is always 0%.

Since the graph is disconnected, the diameter is not well defined and is therefore not evaluated in this regime.

As n increases, the phase transition becomes sharper. For fixed $a > 1$, the size of the largest component grows proportionally with n, while its relative size remains approximately constant. Near the critical point $a \approx 1$, the peak of the second-largest component becomes more pronounced.

1.3 Experiment 2: Logarithmic Regime $p = a \log n/n$

1.3.1 Theoretical Background

Connectivity in random graphs is determined by the disappearance of isolated nodes. When

$$p = \frac{a \log n}{n},$$

the expected number of isolated nodes tends to zero if and only if $a > 1$.

1.3.2 Diameter Estimation

Computing the exact diameter is computationally infeasible for large graphs. We therefore implemented a Double-Sweep BFS heuristic:

- a BFS from an arbitrary node identifies a farthest node u;
- a second BFS from u identifies a farthest node v.

The distance $d(u, v)$ provides a very accurate estimate of the diameter with linear complexity $O(n + |E|)$.

Logarithmic Case Results (Connectivity Threshold)

n	a	Largest (mean)	Second (mean)	Connected (%)	Diameter (mean)
10000	0.9	9997.7	0.8	20.0%	7.0
10000	0.95	9997.9	0.9	10.0%	8.0
10000	0.98	9998.7	0.8	20.0%	7.0
10000	1.0	9999.5	0.4	60.0%	6.67
10000	1.02	9999.1	0.6	40.0%	6.25
10000	1.05	9999.5	0.3	70.0%	6.29
10000	1.1	9999.1	0.6	40.0%	6.25
10000	1.2	10000.0	0.0	100.0%	5.8
100000	0.9	99996.6	1.0	0.0%	N/A
100000	0.95	99997.8	0.8	20.0%	7.5
100000	0.98	99998.8	0.8	20.0%	8.0
100000	1.0	99998.8	0.6	40.0%	7.5
100000	1.02	99999.2	0.6	40.0%	7.25
100000	1.05	99999.2	0.7	30.0%	7.0
100000	1.1	99999.7	0.2	80.0%	7.0
100000	1.2	100000.0	0.0	100.0%	6.3
1000000	0.9	999995.2	1.0	0.0%	N/A
1000000	0.95	999997.2	1.0	0.0%	N/A
1000000	0.98	999998.4	0.8	20.0%	8.0
1000000	1.0	999999.1	0.4	60.0%	7.33
1000000	1.02	999999.0	0.7	30.0%	7.33
1000000	1.05	999999.3	0.6	40.0%	7.0
1000000	1.1	999999.8	0.2	80.0%	7.0
1000000	1.2	1000000.0	0.0	100.0%	7.0

Figure 2: Results for the logarithmic regime ($p = a \log n/n$).

1.3.3 Results and Discussion

In the logarithmic regime, the largest connected component contains almost all nodes, while the second-largest component remains negligible. This indicates that disconnections are caused by a small number of isolated nodes.

As a increases, the probability of full connectivity grows rapidly. For $n = 10^5$ and 10^6, the graph becomes fully connected with high probability for $a \gtrsim 1.1$.

Once connectivity is achieved, the diameter remains very small, typically between 6 and 8, even for $n = 10^6$. This confirms the *Small World* property: although the network is extremely large, the typical distance between nodes grows only logarithmically with the network size.

2 Efficient Generation of Sparse Random Graphs

2.1 Motivation

The naive generation of a random graph with n nodes requires testing all $\binom{n}{2}$ possible pairs of nodes to decide whether an edge is present. This leads to a quadratic time complexity $O(n^2)$. Such an approach is particularly inefficient in the sparse regime, where the edge probability satisfies $p = O(1/n)$ and the expected number of edges is only $O(n)$. As a result, most of the computational effort is spent checking pairs of nodes that are not connected.

2.2 Geometric Skipping Algorithm

To address this inefficiency, we implemented a generator based on *geometric skipping*, which avoids explicitly testing every possible pair of nodes. The key observation is that edges are generated independently with probability p, implying that the number of consecutive absent edges between two present edges follows a geometric distribution with parameter p.

Instead of scanning all pairs sequentially, the algorithm proceeds as follows:

1. All potential edges are conceptually ordered in a one-dimensional sequence corresponding to the upper triangular part of the adjacency matrix.

2. Starting from the first position, the algorithm samples a geometric random variable that represents the number of non-edges to skip.

3. Using this skip length, the algorithm jumps directly to the next edge position and inserts the corresponding edge into the graph.

4. The procedure is repeated until the end of the sequence is reached.

Each iteration of the algorithm produces exactly one edge, and no computation is wasted on examining absent edges. Since the expected number of edges in the sparse regime is linear in n, the overall expected running time of the algorithm is $O(n)$.

2.3 Empirical Verification

To empirically verify the theoretical complexity, we measured the execution time of the algorithm for increasing values of n up to 10^6. Figure 3 shows the execution time on a log–log scale. The measured data closely follow a straight line with slope approximately equal to 1, which is characteristic of linear time complexity. This confirms that the geometric skipping algorithm efficiently generates sparse random graphs in $O(n)$ time.

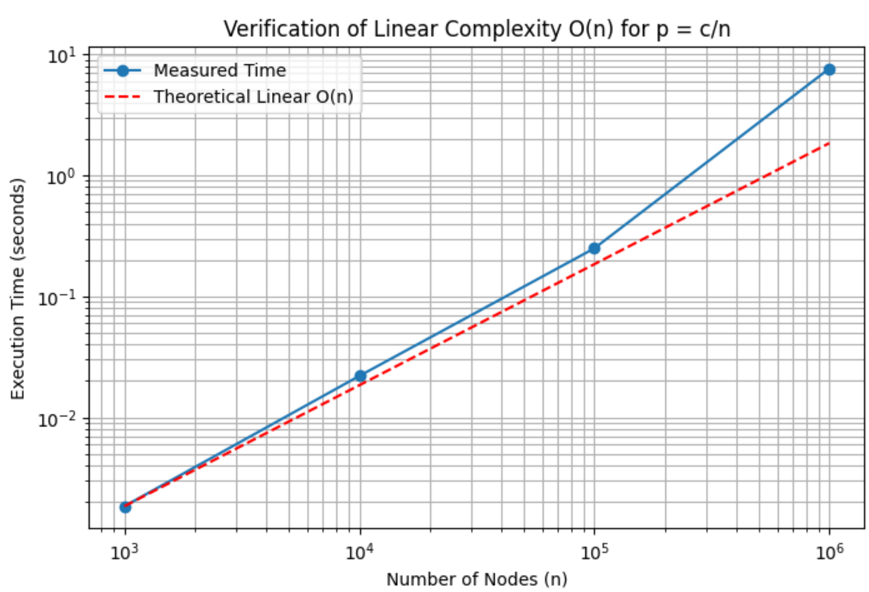

Figure 3: Verification of $O(n)$ complexity.

3 Random Geometric Graphs $G(n, r)$

3.1 Model and Implementation

Nodes are placed uniformly in the unit square $[0, 1]^2$, and edges are added between nodes whose Euclidean distance is less than r. To avoid $O(n^2)$ distance computations, we used a KD-tree spatial index, reducing the complexity to approximately $O(n \log n)$.

3.2 Results and Interpretation

With

$$r = \sqrt{\frac{a \log n}{\pi n}},$$

a connectivity transition is observed near $a = 1$, although it is slower than in non-geometric graphs due to boundary effects.

G(n,r) Geometric Graph Analysis

n	a	Largest (mean)	2nd Largest (mean)	Connected (%)	Diameter (mean)
1000	0.8	864.1	83.8	0.0%	N/A
1000	0.9	961.2	16.7	0.0%	N/A
1000	0.95	970.6	13.3	0.0%	N/A
1000	1.0	974.9	19.3	0.0%	N/A
1000	1.05	990.1	5.6	0.0%	N/A
1000	1.1	996.3	2.0	10.0%	43.0
1000	1.2	996.4	1.9	10.0%	40.0
10000	0.8	9959.3	7.3	0.0%	N/A
10000	0.9	9986.8	5.7	0.0%	N/A
10000	0.95	9986.4	5.8	0.0%	N/A
10000	1.0	9994.4	3.3	10.0%	109.0
10000	1.05	9997.2	1.7	10.0%	102.0
10000	1.1	9996.3	1.6	30.0%	100.0
10000	1.2	9996.5	2.3	30.0%	94.33
100000	0.8	99967.8	6.3	0.0%	N/A
100000	0.9	99984.2	4.6	0.0%	N/A
100000	0.95	99991.3	3.0	0.0%	N/A
100000	1.0	99995.7	1.7	10.0%	294.0
100000	1.05	99995.8	2.0	40.0%	282.5
100000	1.1	99997.4	1.8	10.0%	275.0
100000	1.2	99999.1	0.7	60.0%	259.0
1000000	0.8	999958.4	5.2	0.0%	N/A
1000000	0.9	999987.3	3.3	0.0%	N/A
1000000	0.95	999992.1	2.7	0.0%	N/A
1000000	1.0	999995.6	2.0	20.0%	816.5
1000000	1.05	999998.3	1.1	40.0%	789.5
1000000	1.1	999999.2	0.5	60.0%	767.5
1000000	1.2	999999.4	0.5	60.0%	726.83

Figure 4: Results for random geometric graphs $G(n, r)$.

3.3 Diameter Scaling

When connected, the diameter is very large (hundreds of hops for $n = 10^6$). Since edges are strictly local, long-range shortcuts are absent. The diameter scales as $\sqrt{n/\log n}$, confirming that geometric graphs are *Large Worlds*.

4 Majority Model Simulation

4.1 Model Description

A dynamic majority model was simulated on a random geometric graph $G(n, r)$ with $n = 10^4$ nodes and connection radius

$$r = \sqrt{\frac{3 \log n}{\pi n}},$$

which guarantees connectivity with high probability. Each node holds a binary opinion $\sigma_i(t) \in \{+1, -1\}$ at time t.

The dynamics follow a **synchronous majority update rule**: at each discrete time step, every node updates its opinion by adopting the majority opinion among its neighbors. In case of a tie, the node keeps its current opinion. The process is iterated until a consensus state is reached or a maximum number of iterations is exceeded.

4.2 Experimental Setup

The graph structure and neighbor queries are handled using the `NetworkX` library. At time $t = 0$, each node is initialized independently with opinion $+1$ with probability $p_{\text{init}} \in \{0.5, 0.6, 0.7, 0.8\}$ and with opinion -1 otherwise.

For each value of p_{init}, multiple independent simulations are performed in order to empirically estimate:

- the probability of reaching the consensus state where all nodes have opinion $+1$;
- the average number of time steps required to reach such a consensus;
- the average wall-clock simulation time.

The process is terminated as soon as a unanimous configuration is detected.

4.3 Results and Discussion

The experimental results are summarized in Table 1. When the initial configuration is perfectly balanced ($p_{\text{init}} = 0.5$), the system exhibits strong fluctuations and the final consensus is essentially random. In this regime, convergence is slower and the probability of reaching the $+1$ consensus is close to 50%.

As soon as a small initial bias is introduced ($p_{\text{init}} \geq 0.6$), the dynamics change dramatically. The system converges to the $+1$ consensus in 100% of the simulations and the convergence time drops to only two or three iterations. This shows that the majority rule acts as a powerful nonlinear amplifier: even a modest initial advantage is sufficient to deterministically drive the system toward global agreement.

The extremely fast convergence highlights the strong mixing properties of random geometric graphs at the connectivity threshold. Local interactions are sufficient to propagate the dominant opinion across the entire network in only a few synchronous updates.

Table 1: Majority Model Convergence Results on $G(n, r)$ with $n = 10^4$.

p_{init}	**Prob Consensus (+1)**	**Avg Steps (to +1)**	**Avg Time (s)**
0.5	56.0%	6.46	0.4223
0.6	100.0%	2.64	0.2388
0.7	100.0%	2.00	0.1994
0.8	100.0%	1.94	0.2191

Overall, the simulations confirm that on well-connected geometric graphs, the majority dynamics exhibit a sharp phase transition around $p_{\text{init}} = 0.5$. Above this threshold, consensus on the dominant opinion emerges almost instantaneously and with probability one.

5 Conclusion

This laboratory confirmed key theoretical predictions on random graphs and dynamical processes. We observed phase transitions in connectivity, verified the efficiency of linear graph generation algorithms, highlighted the structural differences between random and geometric graphs, and demonstrated rapid consensus formation in majority dynamics. Several extensions could further enrich the analysis. First, the study could be expanded to larger network sizes in order to investigate finite-size effects and better validate the asymptotic theoretical results. Second, heterogeneous node distributions or spatially correlated placements could be introduced in the geometric graph model to capture more realistic spatial constraints. These extensions would allow for a deeper understanding of the interplay between network structure and dynamical processes and would provide a more comprehensive framework for the study of collective behavior in complex systems.